🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.EE.3

Properties of Operations

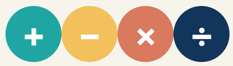
Lesson 6-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary

Level 2 Standard

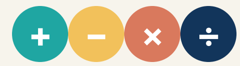
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$3 + 7 = 7 + 3 = 10$ both ways; $4 \times 5 = 5 \times 4 = 20$ both ways — order does not matter

Commutative Property

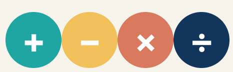
Write the definition:



$(2 + 3) + 4 = 5 + 4 = 9$ and $2 + (3 + 4) = 2 + 7 = 9$ — same answer, different grouping

Associative Property

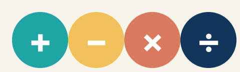
Write the definition:



$9 + 0 = 9$ (zero is the identity for addition); $6 \times 1 = 6$ (one is the identity for multiplication)

Identity Property

Write the definition:



Commutative works for + and ×, but NOT for − or ÷ (since $5 - 3 = 2$ but $3 - 5 = -2$)

Property

Write the definition:

$\underbrace{4}x$

coefficient = 4

In 3x, the 3 is the coefficient — it tells you to multiply x by 3

Coefficient

Write the definition:

$3x + 2x = 5x$

same variable → combine

4x + 2x = 6x (like terms, both have x); 4x + 2y stays as 4x + 2y (unlike terms)

Like terms

Write the definition:

Guided Notes Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can use the commutative, associative, and identity properties to rewrite expressions.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Commutative Property | Associative Property | Identity Property | Property | Coefficient | Like terms

👆 *Tap any word to see what it means and a picture.*

1. The rule that order does not change a sum or product, like a + b = b + a, is the ________________.

2. The rule that grouping does not change a sum or product, like (a + b) + c = a + (b + c), is the ________________.

3. The rule that adding 0 or multiplying by 1 keeps a number the same is the ________________.

4. A rule that is always true in math is a ________________.

5. A number multiplied by a variable is a ________________.

6. Terms with the same variable part are ________________.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Which property is shown? $5 + 13 = 13 + 5$

A. Commutative Property

B. Associative Property

C. Identity Property

D. Distributive Property

Step 1 The order of the addends changed, so this is the Commutative Property of Addition.

✅ **Answer:** A. Commutative Property

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which property is shown? $(6 \times 3) \times 2 = 6 \times (3 \times 2)$

A. Associative Property

B. Commutative Property

C. Identity Property

D. Distributive Property

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which property is shown? $47 + 0 = 47$

A. Identity Property

B. Commutative Property

C. Associative Property

D. Distributive Property

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Track 3 plays $(7 + 3) + 5 = 7 + (3 + 5)$. Only the parentheses moved. Which property is this?

A. Associative Property

B. Commutative Property

C. Distributive Property

D. Identity Property

Show your work:

2. Track 4 plays $1 \times y = y$. Which property is this?

A. Identity Property

B. Distributive Property

C. Commutative Property

D. Associative Property

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Explain why the commutative property works for addition and multiplication but NOT for subtraction and division. Give a specific number example for each operation to prove your point.

Sentence starter: Addition: ____ + ____ = ____ + ____ = ____ (works). Subtraction: ____ − ____ ≠ ____ − ____ because ____. Multiplication: ____ × ____ = ____ × ____ = ____ (works). Division: ____ ÷ ____ ≠ ____ ÷ ____ because ____.

Show your work:

Reflect — Exit Ticket

Which property is shown? (8 + 5) + 2 = 8 + (5 + 2)

A. Associative Property

B. Commutative Property

C. Identity Property

D. Distributive Property

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Commutative Property
2. **Notes 2:** Associative Property
3. **Notes 3:** Identity Property
4. **Notes 4:** Property
5. **Notes 5:** Coefficient
6. **Notes 6:** Like terms
7. **We Try (notes):** A. Associative Property *— The grouping (parentheses) changed but the order stayed the same, so this is the Associative Property of Multiplication.*
8. **You Try (notes):** A. Identity Property *— Adding 0 does not change the value, so this is the Identity Property of Addition.*
9. **Try It 1:** A. Associative Property *— The grouping (parentheses) changed but the order of the numbers stayed the same, so this is the Associative Property.*
10. **Try It 2:** A. Identity Property *— Multiplying by 1 keeps the same value, so this is the Identity Property of Multiplication.*
11. **Exit Ticket:** A. Associative Property *— The grouping (parentheses) changed but the numbers stayed in the same order, so this is the Associative Property of Addition.*